Practice workbook · Describe how blood clots and explain why clotting matters

The sequence from damaged vessel and platelets to a fibrin mesh, labelled flow charts, the benefits of clotting at a wound and when clotting must be blocked.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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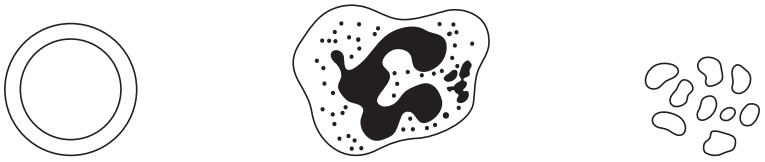
Type ID: HBI-BLOOD-CLOTTING-PROCESS-AND-IMPORTANCE

1. Question 3 · part b.i · 5 marks

November 2021 Paper 1

Worked-example retry · Attempt part b.i only.

3 The diagram shows some of the components of blood.



A B C

(a) Complete the table by giving the name of the component and its function.

Component A has been completed as an example.

(4)

Component	Name	Function
A	red blood cell	transports oxygen
B		
C		

(b) (i) Describe the process of blood clotting.

(5)

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1. Question 3 · part b.i · 5 marks

November 2021 Paper 1

Worked-example retry · Attempt part b.i only.

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(ii) Explain why it is important that blood clots.

(3)

(Total for Question 3 = 12 marks)

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1. Question 3 · part b.i · 5 marks · Working and answer

Working and answer

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2. Question 9 · part a.i · 3 marks

June 2019 Paper 2

Further practice · Attempt part a.i only.

9 Aspirin is a painkiller, but can also be prescribed to people who are at risk of having a stroke or a heart attack.

Aspirin dissolves blood clots and is also an enzyme inhibitor, which reduces the risk of more blood clots forming in blood vessels.

(a) (i) Describe how a blood clot forms in a blood vessel.

(3)

(ii) Explain how a blood clot in the coronary artery increases the risk of a heart attack.

(3)

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2. Question 9 · part a.i · 3 marks · Working and answer

Working and answer

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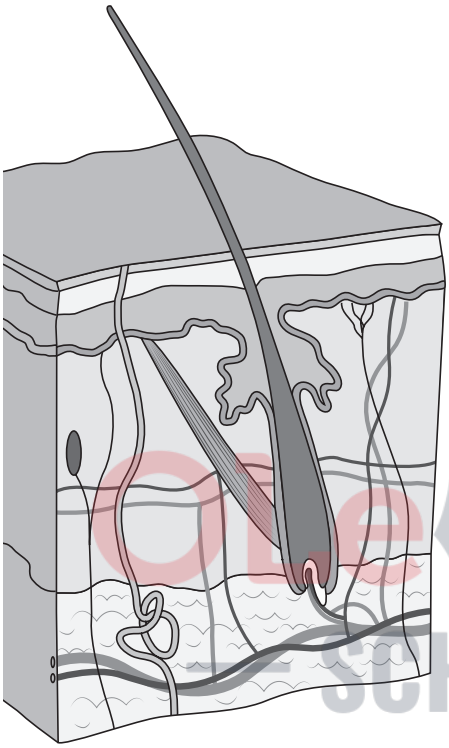
3. Question 5 · part d · 4 marks

January 2020 Paper 2

Further practice · Attempt part d only.

5 Burns can be first degree, second degree or third degree depending on how many layers of the skin are damaged.

The diagram shows the structure of the skin and the layers damaged by each type of burn.



layer 1	first degree burn
layers 1 and 2	second degree burn
layers 1, 2 and 3	third degree burn

(a) Name the three layers of skin that are damaged by a third degree burn.

(3)

layer 1 .....

layer 2 .....

layer 3 .....

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3. Question 5 · part d · 4 marks

January 2020 Paper 2

Further practice · Attempt part d only.

(b) Which two skin structures are damaged by a second degree burn?

(1)

- ☒ **A** blood vessels and sebaceous glands
- ☒ **B** fat tissue and blood vessels
- ☒ **C** hair follicles and sweat glands
- ☒ **D** sebaceous glands and fat tissue

(c) Explain why a person with a second or third degree burn has a high risk of infection.

(2)

(d) Blood clotting helps to prevent infection.

Describe the process of blood clotting.

(4)

(Total for Question 5 = 10 marks)

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3. Question 5 · part d · 4 marks · Working and answer

Working and answer

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4. Question 6 · part a.ii · 4 marks

June 2022 Paper 2R

Further practice · Attempt part a.ii only.

6 Blood will clot when there is damage to the skin, such as a cut.

(a) (i) State two reasons why it is important for the blood to clot.

(2)

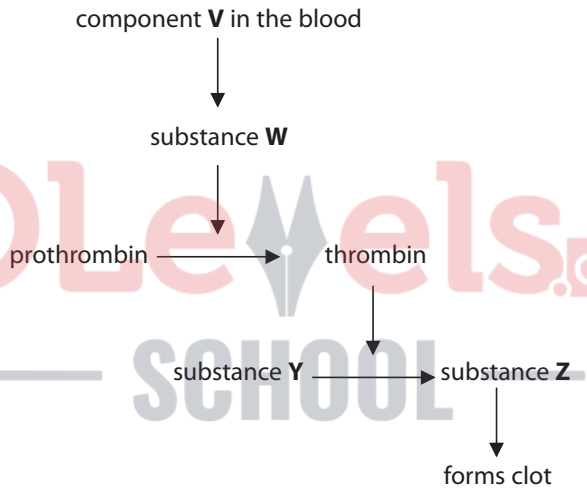
1 .....

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2 .....

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(ii) The diagram shows some of the stages in the process of blood clotting.



Use words from the box to identify V, W, Y and Z.

thrombokinase enzyme		white blood cells		platelets
haemoglobin	fibrinogen	fibrin	antigens	antibody

(4)

V .....

W .....

Y .....

Z .....

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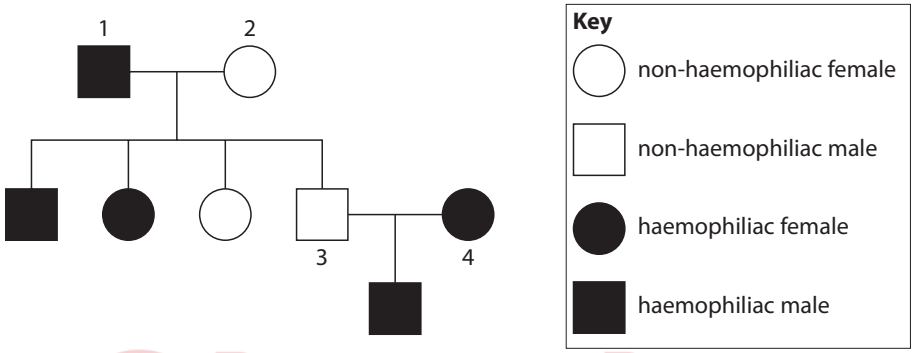
4. Question 6 · part a.ii · 4 marks

June 2022 Paper 2R

Further practice · Attempt part a.ii only.

(b) Haemophilia is an inherited condition where the blood of a person does not clot.

The diagram shows a family pedigree. Some members of the family have the condition.



(i) Deduce the genotypes of family members 1 and 2 showing how you obtained your answer.

(5)

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4. Question 6 · part a.ii · 4 marks · Working and answer

Working and answer

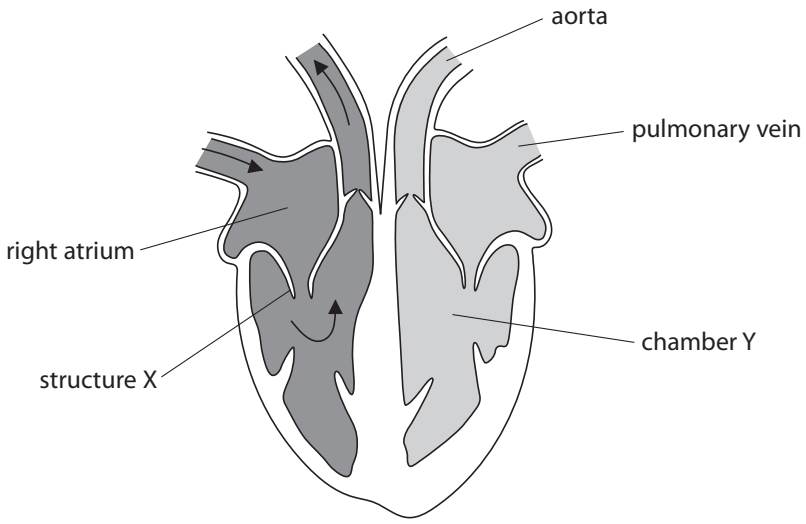
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5. Question 2 · part c.ii · 2 marks

January 2023 Paper 2

Further practice · Attempt part c.ii only.

2 (a) The diagram shows a section through the heart.



(i) Name chamber Y. **(1)**

(ii) Explain the function of structure X. **(2)**

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P 7 2 6 0 2 A 0 4 2 8

5. Question 2 · part c.ii · 2 marks

January 2023 Paper 2

Further practice · Attempt part c.ii only.

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(iii) The table lists some properties of blood.

Add ticks (✓) to those boxes that show the properties of blood in the aorta.

(2)

Properties of blood	Blood in the aorta
oxygenated	
deoxygenated	
high pressure	
medium pressure	
low pressure	

(iv) State where blood in the left side of the heart has come from.

(1)

(b) In one year in the United Kingdom, there were 173 heart transplants.

Heart transplant operations require hearts to be donated.

Some hearts cannot be used because the donor is too old.

Give two other reasons why a heart might not be suitable for a transplant.

(2)

1 .....

.....

2 .....

.....

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5. Question 2 · part c.ii · 2 marks

January 2023 Paper 2

Further practice · Attempt part c.ii only.

(c) (i) Which blood component causes blood clotting?

(1)

- ☒ **A** lymphocytes
- ☒ **B** phagocytes
- ☒ **C** platelets
- ☒ **D** red blood cells

(ii) The diagram summarises part of the process that causes a blood clot.



Anticoagulant drugs prevent a blood clot forming.

These drugs stop the enzyme thrombin being produced.

Describe how a lack of thrombin prevents a blood clot forming.

(2)

(Total for Question 2 = 11 marks)

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5. Question 2 · part c.ii · 2 marks · Working and answer

Working and answer

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6. Question 6 · part a.i · 3 marks

November 2020 Paper 2R

Further practice · Attempt part a.i only.

6 Haemophilia is a condition in which blood does not clot. It is caused by a sex-linked allele.

(a) (i) Explain why the process of blood clotting is important.

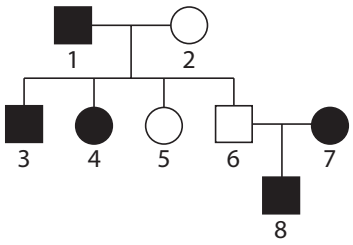
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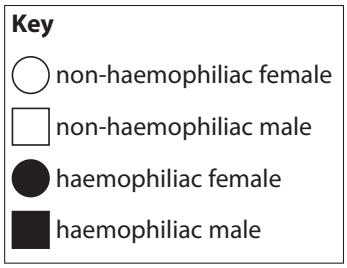
(ii) State what is meant by the term **sex-linked**.

(2)

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(b) The pedigree shows the pattern of inheritance of haemophilia in a family.





X^h is the allele for haemophilia and X^H is the allele for clotting.

(i) What is the genotype of person 3?

(1)

- ☒ **A** $X^h X^h$
- ☒ **B** $X^H X^h$
- ☒ **C** $X^H Y$
- ☒ **D** $X^h Y$

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6. Question 6 · part a.i · 3 marks · Working and answer

Working and answer

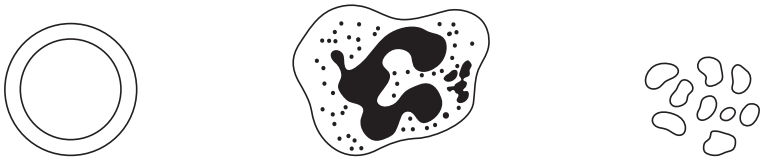
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7. Question 3 · part b.ii · 3 marks

November 2021 Paper 1

Further practice · Attempt part b.ii only.

3 The diagram shows some of the components of blood.



A **B** **C**

(a) Complete the table by giving the name of the component and its function.

Component A has been completed as an example.

(4)

Component	Name	Function
A	red blood cell	transports oxygen
B		
C		

(b) (i) Describe the process of blood clotting.

(5)

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7. Question 3 · part b.ii · 3 marks

November 2021 Paper 1

Further practice · Attempt part b.ii only.

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(ii) Explain why it is important that blood clots.

(3)

(Total for Question 3 = 12 marks)

Working and answer

8. Question 6 · part a.i · 2 marks

June 2022 Paper 2R

Further practice · Attempt part a.i only.

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6 Blood will clot when there is damage to the skin, such as a cut.

(a) (i) State two reasons why it is important for the blood to clot.

(2)

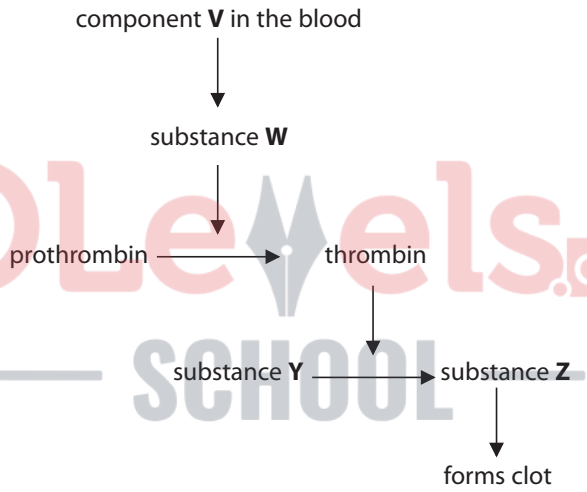
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(ii) The diagram shows some of the stages in the process of blood clotting.



Use words from the box to identify V, W, Y and Z.

thrombokinase enzyme		white blood cells		platelets
haemoglobin	fibrinogen	fibrin	antigens	antibody

(4)

V .....

W .....

Y .....

Z .....

8. Question 6 · part a.i · 2 marks · Working and answer

Working and answer

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9. Question 1 · part b · 4 marks

November 2023 Paper 1

Further practice · Attempt part b only.

1 (a) The table gives components of the blood and their features.

Complete the table by putting a tick (✓) in the box if the blood component shows the feature.

(5)

Feature	Blood component				
	red blood cell	phagocyte	lymphocyte	platelet	plasma
transports oxygen					
contains haemoglobin					
is a fluid					
produces antibodies					
engulfs pathogens					

(b) Explain why it is important that blood clots when the skin is cut open.

(4)

(Total for Question 1 = 9 marks)

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9. Question 1 · part b · 4 marks · Working and answer

Working and answer

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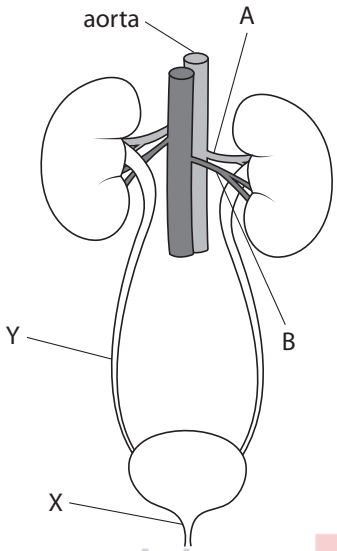
10. Question 8 · part b.ii · 3 marks

January 2020 Paper 1

Further practice · Attempt part b.ii only.

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8 The diagram shows part of the human renal system.



(a) (i) Give the names of structures X and Y.

(2)

structure X .....

structure Y .....

(ii) Describe the function of the aorta.

(2)

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(iii) Compare the composition of the blood in vessels A and B.

(2)

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10. Question 8 · part b.ii · 3 marks

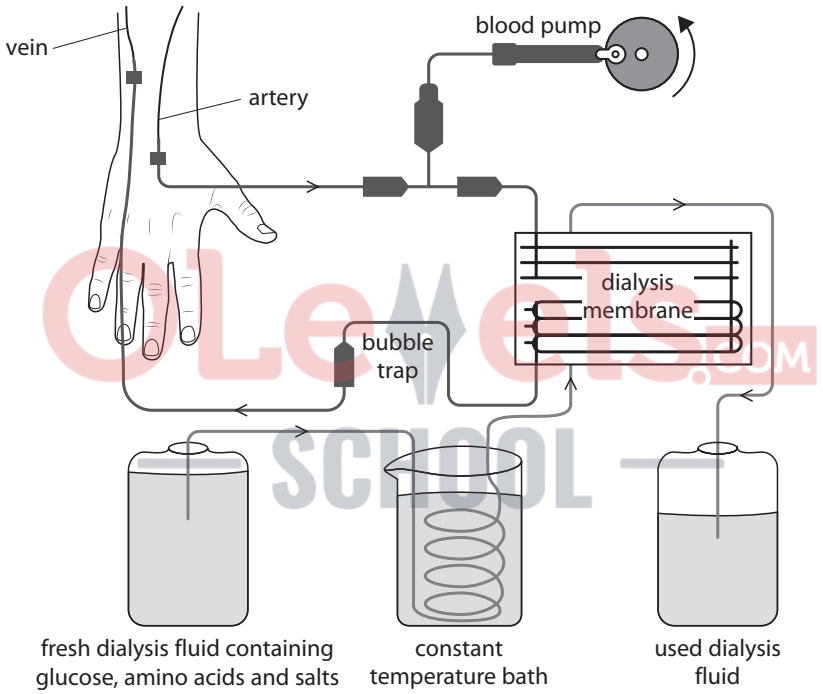
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Further practice · Attempt part b.ii only.

(b) Dialysis is a treatment used for people with kidney failure.

Blood containing waste substances is pumped through a dialysis machine. The dialysis machine filters the waste from the blood. The cleaned blood is then returned to the patient.

The diagram shows how a dialysis machine filters the blood of a person with kidney failure.



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10. Question 8 · part b.ii · 3 marks

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Further practice · Attempt part b.ii only.

(i) Explain how diffusion removes waste substances from the patient's blood inside the dialysis machine.

(4)

(ii) A chemical is added to the patient's blood as it enters the dialysis machine. This chemical prevents the blood from clotting.

Explain why it is important to add this chemical as the blood enters the machine, rather than just before the blood returns to the body.

(3)

(Total for Question 8 = 13 marks)

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10. Question 8 · part b.ii · 3 marks · Working and answer

Working and answer

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